BAUTH

A Keystroke Dynamics Authentication System Using Machine Learning

PROJECT VISION

- The evolution of technology has rendered traditional username and password authentication increasingly vulnerable.
- Multi-factor authentication required for security.
- Human behavior is unique, so is typing behaviour.
- Keystroke dynamics as secondary-factor authentication.
- BAuth is a web-based system that intergrates Machine learning to authenticate users based on their keystroke dynamics.

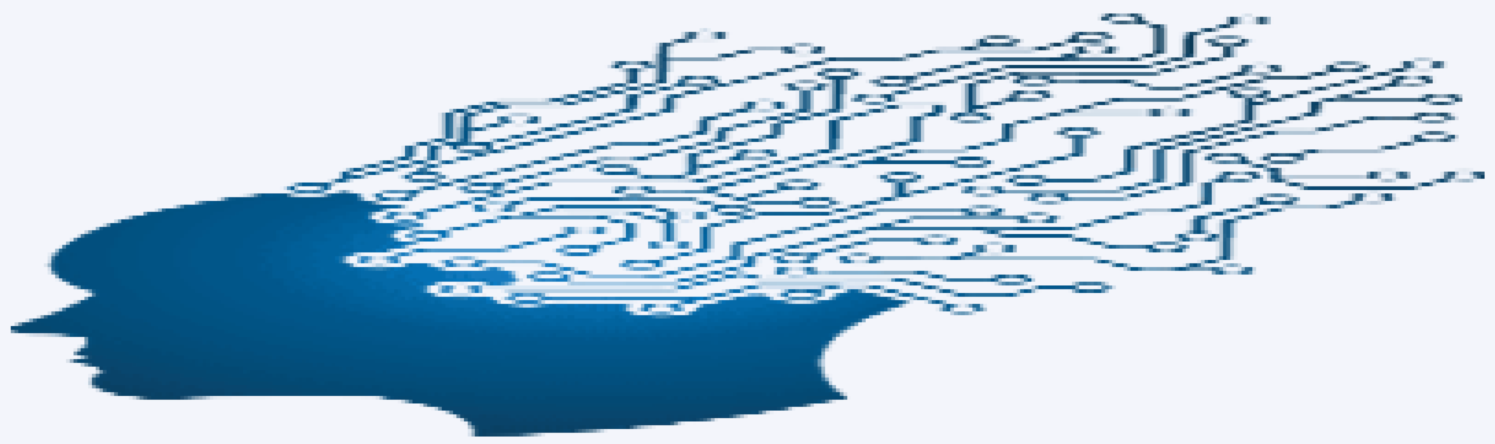
PERFORMANCE:

- 51 Users
- 400 sessions each
- 20,400 data rows

ACCURACY SCORE

>0.93

Increases with data

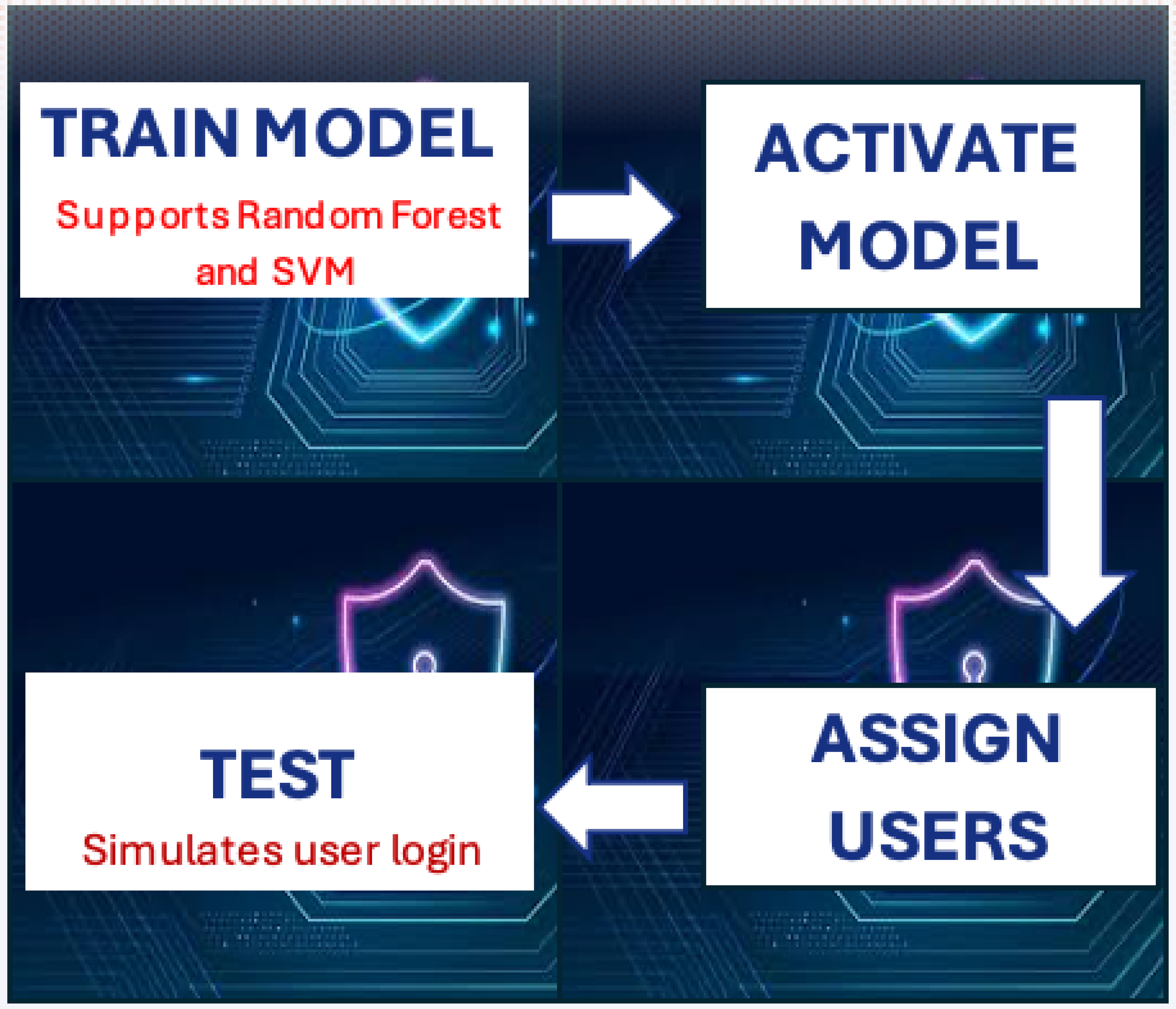


TECHNOLOGIES

- Machine Learning
- MySQL
- Python
- Flask
- HTML
- PHP
- API
- JavaScript
- CSS

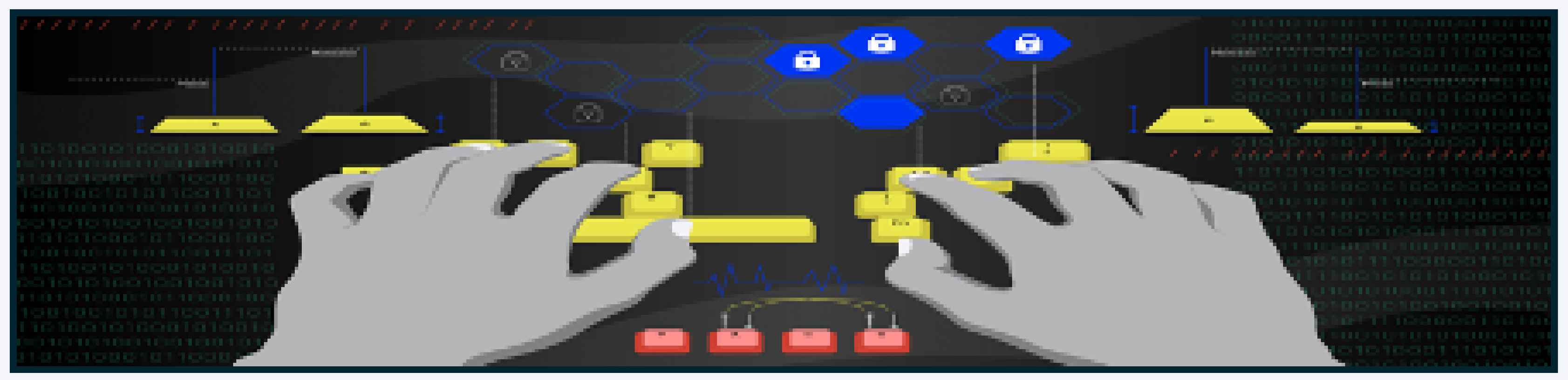
BENEFITS OF BAUTH

- **Enhanced Security:** Reduces the risk of credential theft and unauthorized access.
- **User Convenience:** Eliminates the need for additional devices like OTP tokens.
- **Cost-Effective:** No additional hardware required, making it cost-efficient.
- **Scalability:** Can be integrated into various authentication systems with minimal modifications.
- **Continuous Improvement:** Machine learning models continuously learn and improve authentication accuracy over time.



KEY FUNCTIONALITIES

- **Users:** Controls user information.
- **Model:** Handles model training. Supports Random Forest and SVM models. View model performances over time.
- **Authentication:** Manages user's authentication methods. Username and password as default. Keystroke as optional secondary authentication method.
- **Test:** Simulates a user login page where the admin tests the performance of the trained model.



https://github.com/Bukka101/BAuth-keystroke-authentication-system-using-machine-learning-by-stephen-nzekwesi